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PHASE 2

PROJECT PROPOSAL

HEPHAESTUS

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1.0 Introduction

Hasta Travel Car Rental System is a proposed database driven web-based information system. It is designed to enhance vehicle booking experience and allow management of vehicle related services. The current system deeply relies on manual staff involved activities to deal a booking. Thus, the proposed system aims to be automated for booking services with real-time vehicle availability tracking as well as deliver a more convenient and user-friendly booking experience for customers. There will also be automated invoices and integration with e-invoicing through the Inland Revenue Board of Malaysia (IRBM) to reduce manual workload and minimize errors.

The main focus for current phase of project proposal is to identify system requirements and propose a suitable database design, including a To-Be Data Flow Diagram (DFD), data and transaction requirements, conceptual ERD, enhanced ERD (EERD) and a data dictionary. This proposal will serves as a foundation for developing a reliable and scalable database system that supports the functional requirements of the Hasta system and aligns with good database design principles.

2.0 DFD (to-be)

The Data Flow Diagram (DFD) illustrates how data flows within the proposed Hasta Travel car rental system and provides a clear overview of interactions between users, processes and data stores. To-Be DFD serves as a visual representation of the system's functional requirements and supports a better understanding of how the proposed system operates in an integrated and structured manner.

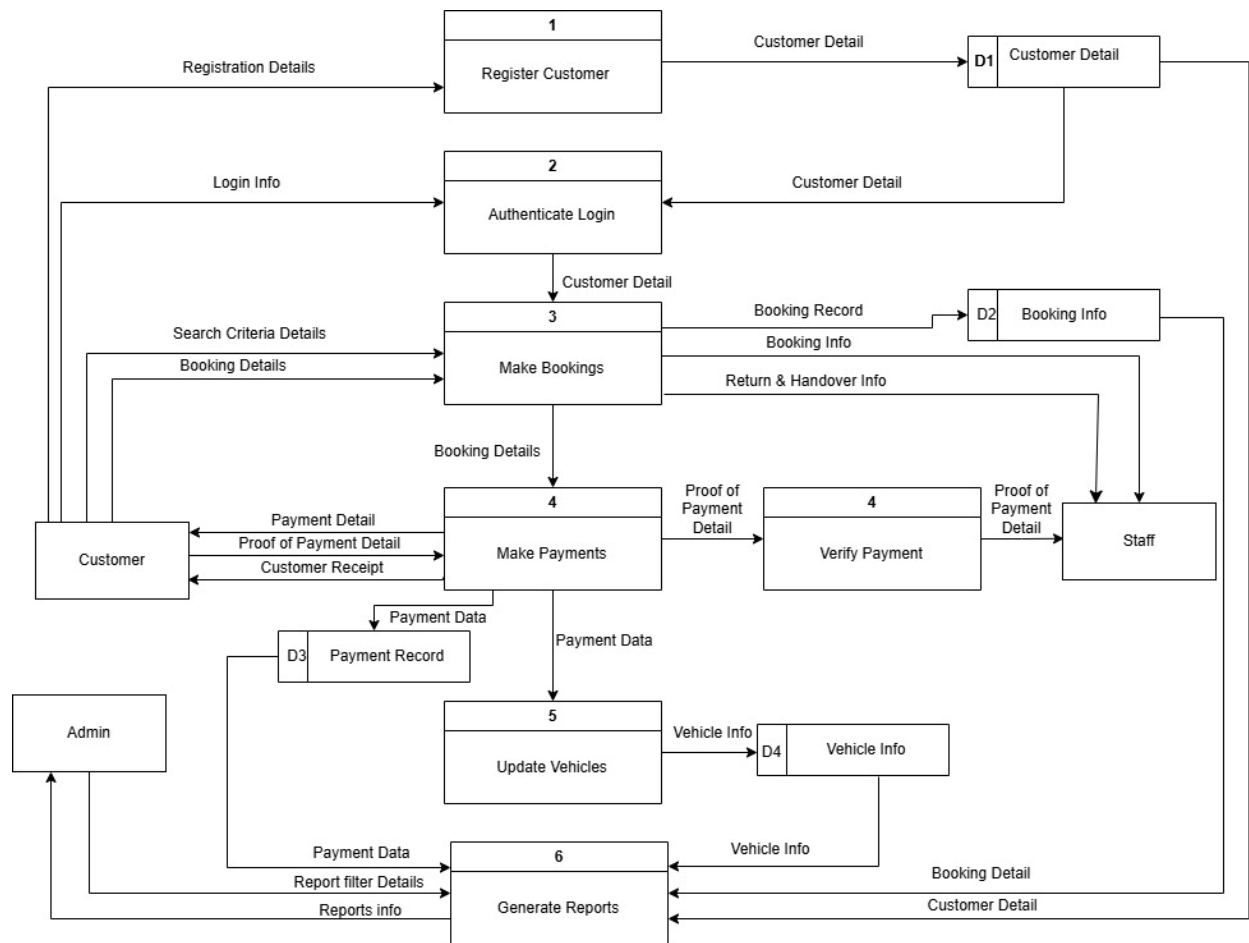


Figure 1: Level 0 Logical DFD

3.0 Data & Transaction requirement

This section defines the key data rules and transaction constraints to ensure Hasta Travel Car Rental System operations (booking, payment verification, return, cancellation, and refund) are recorded consistently and accurately in the database.

3.1 Proposed business rule

Customer Registration & Login: Each customer must register an account before making a booking. Registration requires valid customer details and is used to securely access booking history and account functions.

Vehicle Browsing & Availability: Customers may browse vehicles without logging in, but the system must display only vehicles that are available for the selected rental period to prevent double-booking.

Booking Creation: Each booking must have a unique booking number and must be linked to exactly one customer and one vehicle. A customer can create multiple bookings over time.

Booking Status Control: Bookings are tracked using statuses (e.g., Pending, Approved, Rejected, Cancelled, Completed). Status changes must be recorded to support tracking and auditing.

Two-Stage Payment Requirement (Deposit + Rental Fee): Customers must make two payments for each booking: (1) deposit and (2) rental fee. Each payment must be recorded with date/time, amount, and uploaded receipt for verification.

Payment Verification by Staff: All submitted payments must be verified by staff. Staff actions (approve/reject/request resubmission) and verification remarks must be stored and must update booking/payment status accordingly.

Multiple Payment Transactions per Booking: A booking may have multiple payment transactions such as deposit payment, rental payment, penalty charges, and deposit refunds. Each transaction must have its own payment record for traceability.

Cancellation & Deposit Rule: If a customer cancels more than 24 hours before the rental time, the deposit is refundable. If cancellation is within 24 hours, the deposit is forfeited (no refund).

Car Pickup & Return Completion Rule: A booking is considered complete only after the car is returned and all related payments (including penalties if any) are settled and verified.

Deposit Refund Rule (After Return): After the car is returned, the deposit refund must be recorded as full, partial, or withheld depending on return condition (e.g., damages or late return). Refund amount and refund date must be stored.

Vehicle Availability Update: Once a booking is completed and the vehicle is returned, the system must update the vehicle availability status to “Available” so it can be booked again.

Invoice & Receipt Generation: After successful payment verification, the system generates and stores invoices/receipts (invoice number, booking reference, customer info, payment details, issue date) for future reference and disputes.

Loyalty Reward Rule: Each successful (completed) booking will trigger loyalty stamp/points accumulation for the customer, which can later be used for rewards.

3.2 Proposed data & transactional

AdditionalCharges

The AdditionalCharges entity stores extra charges incurred during a booking, including add-on charges, late return fees, and damage fees. These records support accurate billing and invoice generation.

Booking

The Booking entity stores booking ID, customer ID, vehicle ID, booking creation date, rental start and end dates, pickup and return points, booking status (e.g., Pending, Approved, Rejected, Cancelled, Completed), remarks, and total payable amount. Booking records maintain the full lifecycle of each reservation and link customers, vehicles, and payments.

Customer

The Customer entity stores registered user information such as customer ID, full name, matric number (if applicable), IC/passport number, phone number, email, faculty/college, customer type, registration date, and driving license details. These records are used for authentication, eligibility verification, booking ownership, and communication.

Invoice

The Invoice entity stores invoice ID, booking ID, staff ID, issue date, invoice number, and total amount. Invoices are generated after payment verification to provide official billing records and support reporting.

LoyaltyCard

The LoyaltyCard entity stores loyalty card ID, customer ID, total stamps/points, and last updated date. Loyalty data tracks rewards earned from completed bookings and supports reward redemption.

OwnerCar

The OwnerCar entity stores owner ID, name, IC number, contact information, email, bank name, bank account number, and registration date. Owner details are used when vehicles are associated with external owners and for deposit/refund settlement.

Payment

The Payment entity stores payment ID, booking ID, payment purpose (Deposit, Rental Fee, Penalty, Refund), payment method/type, amount, payment date/time, proof/receipt URL, and payment status (Pending, Approved, Rejected). Staff verification information such as staff ID, verification time, and remarks is also recorded to ensure an audit trail.

Review

The Review entity stores review ID, booking ID and/or vehicle ID, customer ID, rating, comment, and review date. Reviews support customer feedback and quality improvement.

Staff

The Staff entity stores staff ID, name, role/position, branch/location, contact information, login credentials, and account status. Staff records support operational tasks such as booking approval, payment verification, and generating invoices.

SystemLog

SystemLog stores log ID, user type, action description, and timestamp. Logs support auditing, troubleshooting, and accountability for critical system operations.

Vehicle

The Vehicle entity stores vehicle ID, vehicle number, model, type, plate number, seating capacity, transmission, color, daily rental price, availability status, and created date. Vehicle data supports browsing, availability checking, and fleet management.

VehicleDocument

VehicleDocument stores documents related to each vehicle such as document ID, document type (insurance, road tax), upload and verification dates, policy/road tax numbers, expiry dates, and file URLs/images. It supports compliance tracking and verification by staff.

VehicleMaintenance

VehicleMaintenance stores maintenance ID, vehicle ID, mileage, service date, service type, next due date, remarks, and cost. Maintenance records ensure vehicles are serviced on time and help manage fleet readiness.

Voucher

The Voucher entity stores voucher ID, discount type/value, eligibility rules, and active status. Voucher data supports promotional discounts during booking and allows the system to validate voucher availability.

WalletAccount

WalletAccount stores wallet account ID, customer ID, balances (virtual balance, hold amount, available balance), wallet status, and created date. It supports wallet-based payment and refund management (if enabled).

3.2.2 Proposed Transactional Requirement

Data Entry

1. Enter new customer registration details such as full name, IC/passport number, phone number, email, address, and driving license information. (Customer)
2. Enter vehicle search criteria such as vehicle model, type, seating capacity, transmission, and rental period to check availability. (Customer)
3. Create a new booking by entering pickup point, return point, rental start and end dates, selected vehicle, and any voucher applied. (Customer)
4. Record payment submission for a booking (deposit or rental fee) including payment method, amount, payment date/time, and upload payment receipt/proof. (Customer)
5. Enter staff details such as name, role/position, branch, and contact information for staff account creation. (Admin)
6. Enter or register a new vehicle record including model, type, plate number, seating capacity, rental price, and availability status. (Staff/Admin)
7. Upload vehicle documents (insurance, road tax) with policy numbers, expiry dates, and document file URLs for compliance records. (Staff)
8. Record vehicle maintenance activities including service date, service type, mileage, cost, and next due date. (Staff)
9. Generate an invoice after successful payment verification with invoice number, issue date, and total amount. (System/Staff)
10. Create system notifications (booking confirmation, cancellation result, payment status update) and store the notification log. (System)

Data Update/Delete

1. Update or delete customer profile details (phone number, email, address, license information). (Customer/Staff)
2. Update or delete booking details when changes are approved (rental dates, pickup/return point, vehicle reassignment). (Staff)
3. Update booking status (Pending to Approved/Rejected, Approved to Completed, Approved to Cancelled) including staff remarks. (Staff)
4. Update or delete staff account information and account status (active/suspended). (Admin)
5. Update or delete vehicle details (rental price, status, images) and set vehicle availability status (available, reserved, rented, maintenance). (Staff)
6. Update payment status from Pending to Approved/Rejected, and record verification staff ID/time/remarks. (Staff)
7. Update refund-related payment records (deposit refunded, partial refund, penalty deduction) after vehicle return inspection. (Staff/System)
8. Update vehicle document verification status/date and renew document records when insurance/road tax changes. (Staff)
9. Update or delete vehicle maintenance records when services are edited or corrected. (Staff)
10. Activate/deactivate vouchers and update voucher conditions based on promotion validity. (Admin)

Data Queries

1. List the details of all customers, including faculty/college and contact information. (Staff/Admin)
2. Identify the total number of customers registered in the system. (Admin)
3. List all bookings for a specific customer, ordered by booking date. (Customer/Staff)
4. List all bookings in a given month and summarize by status (Pending, Approved, Cancelled, Completed). (Admin)
5. Identify the total number of successful bookings within a month. (Admin)
6. List all bookings associated with a specific vehicle and identify the number of bookings for that vehicle. (Staff/Admin)
7. List all vehicles currently available for booking within a selected date range. (Customer/Staff)
8. Identify total payment amount collected in each month, separated by payment purpose (deposit, rental fee, penalty, refund). (Admin)
9. Retrieve payment history for a booking, including verification status and timestamps. (Customer/Staff)
10. Retrieve vehicles due for maintenance or document renewal based on next due date or expiry date. (Staff/Admin)

4.0 Database conceptual design

This section outlines the plan for the Hasta Travel database. It consists of a conceptual ERD and Enhanced ERD model where it focuses on what data needs to be stored. The design uses diagram to show the main entities in the system and how they are related to each other

4.1 Conceptual ERD

Figure 4.1.1 shows the entities that are identified as the core for the system to be functional. Each of the entities has relationships with others that are represented as lines. The attributes of each related entity are identified and described by actual requirement stored data in the system.

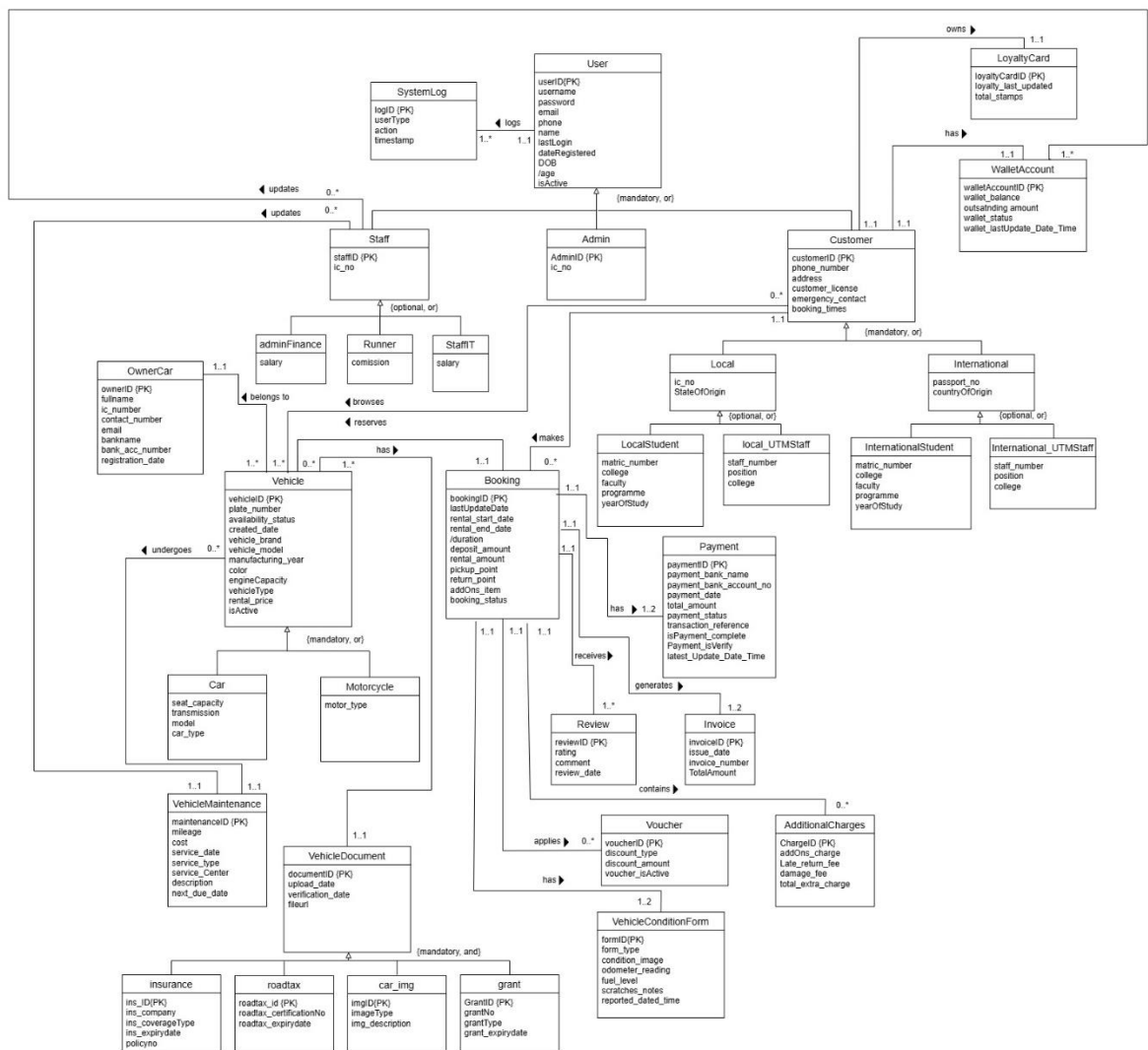


Figure 4.1.1 Conceptual ERD Hasta Travel Car Rental System

4.2 Enhanced ERD (EERD)

Figure 4.2.1 shows EERD, which is more detailed in the version of the conceptual ERD. The concept of EERD is to include superclass and subclass for entities by implementing Generalization/Specialization. This approach allows entities that share common attributes to be grouped under a superclass, while attributes that differ are organized into appropriate subclasses, resulting in a clearer and more structured data model. The two constraints, participation and disjoints were applied to show the relationship between superclass and its subclasses. For example, Customer is superclass to the LocalStudent, UTMStaff and InternationalStudent, with the constraint of Mandatory and Or. It is defined that the Customer must be one of the three subclasses, and Customer cannot occur in two or three subclasses at the same time.

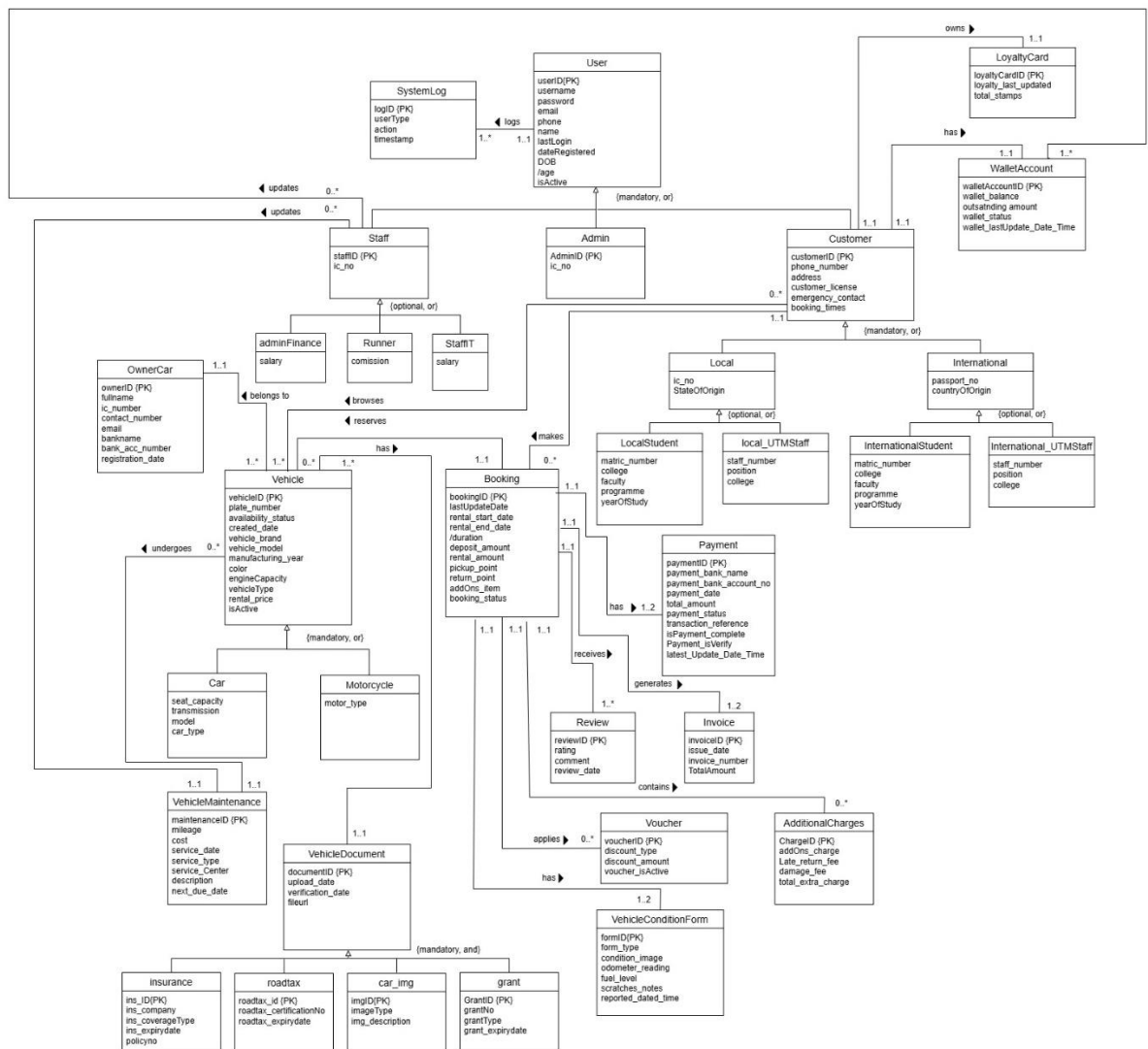


Figure 4.2.1 EERD for Hasta Travel Car Rental System

5.0 Data dictionary

This section provides a listing of the major system of entities in the Hasta Car Rental System. Each entity represents a domain-level class identified in the domain model. The table describes the purpose of each class and its key attributes. These classes are used to store and manage system information and are implemented as tables in the database.

5.1 Entity: <AdditionalCharges>

ATTRIBUTE NAME	TYPE	DESCRIPTION
ChargeID (PK)	UUID	Unique Identifier for each additional charge
addOns_charge	DECIMAL	Additional charges for optional add-ons selected during rental
Late_return_fee	DECIMAL	Fee imposed when the vehicle is returned later than the agreed time
damage_fee	DECIMAL	Charge applied for damages identified after vehicle return
total_extra_charge	DECIMAL	Total amount of extra charges calculated from add-ons, late return, and damage fees

5.2 Entity: <Admin>

ATTRIBUTE NAME	TYPE	DESCRIPTION
AdminID (PK)	UUID	Unique Identifier for Admin
ic_no	INT	Identification card number of Admin

5.3 Entity: <AdminFinance>

ATTRIBUTE NAME	TYPE	DESCRIPTION
salary	DECIMAL	The monthly base salary of the staff member.

5.4 Entity: <Booking>

Attribute Name	TYPE	DESCRIPTION
bookingID (PK)	UUID	Unique booking identifier. PK
lastUpdateDate	DateTime	Date and time when the booking record was last updated.
rental_start_date	DateTime	Start date of rental
rental_end_date	DateTime	End date of rental
/duration	INT	Derived attribute that represents the total rental period (in days), calculated from rental_start_date and rental_end_date.
deposit_amount	DECIMAL	Amount of deposit paid by the customer to confirm the booking.
rental_amount	DECIMAL	Total rental cost for the booking excluding deposit.
pickup_point	VARCHAR	Pickup location
return_point	VARCHAR	Return location
addOns_item	VARCHAR	Additional items rented
booking_status	VARCHAR	Status of booking

5.5 Entity: <Car>

ATTRIBUTE NAME	TYPE	DESCRIPTION
seating_capacity	INT	Number of seats
transmission	VARCHAR	Auto or Manual
model	VARCHAR	The specific model name of the car
car_type	VARCHAR	The category of the car body

5.6 Entity: <car_img>

ATTRIBUTE NAME	TYPE	DESCRIPTION
imgID{PK}	VARCHAR	Unique Identifier for car's image
imageType	VARCHAR	Type of image from vehicle (interior,back,front,side)
img_description	VARCHAR	Description of vehicle's image

5.7 Entity: <Customer>

ATTRIBUTE NAME	TYPE	DESCRIPTION
customerID (PK)	UUID	Unique identifier for each customer in the system.
phone_number	VARCHAR	The customer's primary mobile phone number for contact.
address	VARCHAR	Residential or mailing address of the customer.
customer_license	VARCHAR	The driving license number used to verify eligibility to drive.
emergency_contact	VARCHAR	Contact details of a person to be notified in case of emergency.
booking_times	DATETIME	Date and time when the customer made a booking.

5.8 Entity: <grant>

ATTRIBUTE NAME	TYPE	DESCRIPTION
GrantID(PK)	UUID	Unique identifier for each grant
grantNo	VARCHAR	Vehicle's grant number
grantType	VARCHAR	Type of grant
grant_expirydate	DATETIME	Expiry date of grant

5.9 Entity: <insurance>

ATTRIBUTE NAME	TYPE	DESCRIPTION
Ins_ID (PK)	UUID	Unique identifier for each insurance record (Primary Key).
ins_company	VARCHAR	The name of the insurance provider for the vehicle.
Ins_coverageType	VARCHAR	Type of insurance coverage
ins_expirydate	DATE	The expiration date of the vehicle's insurance policy.
policyno	VARCHAR	The unique policy number associated with the vehicle's insurance.

5.10 Entity: <International>

ATTRIBUTE NAME	TYPE	DESCRIPTION
passport_no	VARCHAR	International customer's passport number
countryOfOrigin	VARCHAR	Original country of international customer

5.11 Entity: <International_UTMStaff>

ATTRIBUTE NAME	TYPE	DESCRIPTION
staff_number	VARCHAR	Staff's number identification
position	VARCHAR	Position of staff
college	VARCHAR	Name of college staff stays at.

5.12 Entity: <InternationalStudent>

ATTRIBUTE NAME	TYPE	DESCRIPTION
matric_number	VARCHAR	The student identification number
college	VARCHAR	The name of the residential college
faculty	VARCHAR	The academic faculty the student belongs
programme	VARCHAR	Name of the academic programme or course of study.
yearOfStudy	INT	Current academic year or level of the student

5.13 Entity: <Invoice>

ATTRIBUTE NAME	TYPE	DESCRIPTION
invoiceID (PK)	UUID	Unique identifier for each invoice.
issue_date	DATE	Date when the invoice was issued.
invoice_number	VARCHAR	Human-readable invoice number shown to the customer.
Total_amount	DECIMAL	Total amount charged on the invoice.

5.14 Entity: <Local>

ATTRIBUTE NAME	TYPE	DESCRIPTION
ic_no	INT	Identification card number for local
StateOfOrigin	VARCHAR	The state local live at.

5.15 Entity: <LocalStudent>

ATTRIBUTE NAME	TYPE	DESCRIPTION
matric_number	VARCHAR	The student identification number
college	VARCHAR	The name of the residential college
faculty	VARCHAR	The academic faculty the student belongs to
programme	VARCHAR	Local Student's enrolled programme
yearOfStudy	DATE	Student's year of study.

5.16 Entity: <local_UTMStaff>

ATTRIBUTE NAME	TYPE	DESCRIPTION
staff_number	VARCHAR	The unique employee ID for staff members of the rental system.
college	VARCHAR	The name of the residential college
position	VARCHAR	Staff's position

5.17 Entity: <LoyaltyCard>

ATTRIBUTE NAME	TYPE	DESCRIPTION
loyaltyCardID (PK)	INT	Unique identifier for the loyalty card record.
total_stamps	INT	The current count of accumulated stamps or points earned from completed bookings.
Loyalty_last_updated	DATE	The date when the stamp count was last modified (used to track activity or expiration).

5.18 Entity: <Motorcycle>

ATTRIBUTE NAME	TYPE	DESCRIPTION
motor_type	VARCHAR	The engine or fuel source

5.19 Entity: <OwnerCar>

ATTRIBUTE NAME	TYPE	DESCRIPTION
ownerID (PK)	UUID	Unique identifier for each owner in the system.
fullname	VARCHAR	Full name of the vehicle owner.
ic_number	VARCHAR	Identification card number of the owner.
contact_number	VARCHAR	Contact phone number of the owner.
email	VARCHAR	Email address of the owner.
bankname	VARCHAR	Name of the bank used by the owner.
bank_acc_number	VARCHAR	Bank account number of the owner.
registration_date	DATE	Date when the owner was registered in the system

5.20 Entity: <Review>

ATTRIBUTE NAME	TYPE	DESCRIPTION
reviewID (PK)	UUID	Unique identifier for each review record.
rating	INT	Numerical rating given by the user
comment	TEXT	Written feedback or remarks provided by the user.
review_date	DATE	Date when the review was submitted

5.21 Entity: <roadtax>

ATTRIBUTE NAME	TYPE	DESCRIPTION
roadtax_id{PK}	UUID	Unique identifier for roadtax
roadtax_expirydate	DATE	The date when the current road tax for the vehicle expires.
roadtax_certificationNo	VARCHAR	Roadtax's certification number

5.22 Entity: <Runner>

ATTRIBUTE NAME	TYPE	DESCRIPTION
comission	DECIMAL	The extra earnings/bonus paid to staff per successful booking.

5.23 Entity: <Payment>

ATTRIBUTE NAME	TYPE	DESCRIPTION
paymentID	INT	Unique identifier for each payment.
payment_bank_name	VARCHAR	Name of the bank used to make the payment.
payment_bank_account_no	INT	Bank account number from which the payment was made.
payment_date	DATETIME	Date and time when the payment was made.
total_amount	DECIMAL	Total amount paid for the booking, including deposit or full payment.
payment_status	VARCHAR	Current status of the payment
transaction_reference	VARCHAR	Reference number provided by the bank or payment gateway for tracking the transaction.
isPayment_complete	BOOLEAN	Indicates whether the payment process has been completed
Payment_isVerify	BOOLEAN	Indicates whether the payment has been verified by the system or administrator.
latest_Update_Date_Time	DATETIME	Date and time when the payment record was last updated.

5.24 Entity: <Staff>

ATTRIBUTE NAME	TYPE	DESCRIPTION
staffID (PK)	UUID	Unique identifier for each staff member in the system.
ic_no	VARCHAR	Identification card number of the staff member.

5.25 Entity: <StaffIT>

ATTRIBUTE NAME	TYPE	DESCRIPTION
salary	DECIMAL	Monthly or annual salary of the staff/employee. Stores the amount in currency format.

5.26 Entity: <SystemLog>

ATTRIBUTE NAME	TYPE	DESCRIPTION
logID (PK)	INT	Unique identifier for each log record in the system.
userType	VARCHAR	Type of user who performed the action
action	TEXT	Description of the action performed in the system.
timestamp	DATETIME	Date and time when the action was recorded.

5.27 Entity: <Vehicle>

ATTRIBUTE NAME	TYPE	DESCRIPTION
vehicleID (PK)	UUID	Unique vehicle identifier
plate_number	VARCHAR	Vehicle's plate number
availability_status	VARCHAR	Availability status (available, rented, maintenance, reserved)
created_date	DATE	Date vehicle was added
Vehicle_brand	VARCHAR	Brand of vehicle
vehicle_model	VARCHAR	Model of vehicle
Manufacturing_year	INT	The year in which the vehicle was manufactured. Helps track vehicle age for maintenance and valuation.
color	VARCHAR	Vehicle Color
engineCapacity	DECIMAL	Engine displacement or capacity in liters
vehicleType	VARCHAR	Type of vehicle
rental_price	DECIMAL	Daily rental price
isActive	BOOLEAN	Vehicle's validity by staff after add vehicle in system.

5.28 Entity: <VehicleConditionForm>

ATTRIBUTE NAME	TYPE	DESCRIPTION
formID (PK)	UUID	Unique report identifier. PK
form_type	VARCHAR	Type of report (pickup/return)
condition_image	VARCHAR	URL or path to condition image
odometer_reading	INT	Odometer reading
fuel_level	INT	Fuel level percentage
scratches_notes	TEXT	Notes on scratches/damage
reported_date_time	TIMESTAMP	Date and time of report

5.29 Entity: <VehicleDocument>

ATTRIBUTE NAME	TYPE	DESCRIPTION
documentID (PK)	UUID	Unique document identifier
upload_date	DATE	Date document was uploaded
verification_date	DATE	Date document was verified
fileURL	VARCHAR	URL to document file

5.30 Entity: <VehicleMaintenance>

ATTRIBUTE NAME	TYPE	DESCRIPTION
maintenanceID (PK)	UUID	Unique maintenance record identifier. PK
mileage	INT	Mileage at service time
service_date	DATE	Date of service
service_type	VARCHAR	Type of service performed
next_due_date	DATE	Next service due_date
cost	DECIMAL	Cost of maintenance
service_Center	VARCHAR	Name of the workshop or service center that performed the maintenance.
description	TEXT	Additional details or notes about the maintenance performed.

5.31 Entity: <Voucher>

ATTRIBUTE NAME	TYPE	DESCRIPTION
voucherID (PK)	UUID	Unique identifier for the specific voucher instance.
discount_type	VARCHAR	Describes the nature of the reward
discount_amount	DECIMAL	Value of the discount applied based on the discount type.
voucher_isActive	BOOLEAN	A status flag indicating if the voucher is currently valid for use

5.32 Entity: <WalletAccount>

ATTRIBUTE NAME	TYPE	DESCRIPTION
walletAccountID (PK)	UUID	Unique identifier for each wallet account.
wallet_balance	DECIMAL	Current available balance stored in the wallet.
outstanding_amount	DECIMAL	Amount that is still unpaid or pending
Wallet_status	DECIMAL	Current status of the wallet
Wallet_lastUpdate_Date_Time	VARCHAR	Date and time when the wallet account was last updated.

6.0 Summary

The Hasta Travel Car Rental System presents a structured approach to transforming system requirements into a well-defined database design. The proposed To-Be Data Flow Diagram illustrates how data moves through the system, while the identified business rules and transaction requirements ensure that booking, vehicle management, and related operations are handled accurately and consistently.

The conceptual ERD and Enhanced ERD show the main data entities and their relationships needed to manage customers, vehicles, bookings, and payments, allowing the system to support automated booking and accurate vehicle availability. In addition, the data dictionary provides clear definitions of all data elements used in the system.

Overall, this phase establishes a solid database foundation that supports system automation, reduces manual dependency, and prepares the Hasta Travel Car Rental System for reliable implementation and future expansion.